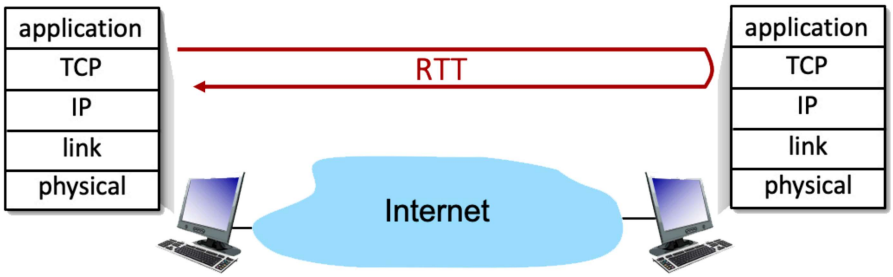
INTERACTIVE END-OF-CHAPTER EXERCISES

COMPUTING TCP'S RTT AND TIMEOUT VALUES

Suppose that TCP's current estimated values for the round trip time (*estimatedRTT*) and deviation in the RTT (*DevRTT*) are 400 msec and 39 msec, respectively (see Section 3.5.3 for a discussion of these variables). Suppose that the next three measured values of the RTT are 350 msec, 340 msec, and 210 msec respectively.



Compute TCP's new value of *DevRTT, estimatedRTT,* and the TCP timeout value after each of these three measured RTT values is obtained. Use the values of α = 0.125, and β = 0.25. Round your answers to two decimal places after leading zeros.

QUESTION LIST

1. What is the estimatedRTT after the first RTT?

2. What is the RTT Deviation for the the first RTT?

3. What is the TCP timeout for the first RTT?

4. What is the estimatedRTT after the second RTT?

5. What is the RTT Deviation for the the second RTT?

6. What is the TCP timeout for the second RTT?

7. What is the estimatedRTT after the third RTT?

8. What is the RTT Deviation for the the third RTT?

9. What is the TCP timeout for the third RTT?

SOLUTION

DevRTT is calculated with the following equation: (1-beta)*DevRTT + beta * |estimatedRTT - sampleRTT|
estimatedRTT is calculated with the following equation: (1-alpha)*estimatedRTT + alpha*sampleRTT
TCP timeout is calculated with the following equation: estimatedRTT + (4*DevRTT)

1. The estimatedRTT for RTT1 is 393.75

2. The DevRTT for RTT1 is 41.75

3. The timeout for RTT1 is 560.75

4. The estimatedRTT for RTT2 is 387.03

5. The DevRTT for RTT2 is 44.75

6. The timeout for RTT2 is 566.03

7. The estimatedRTT for RTT3 is 364.9

8. The DevRTT for RTT3 is 77.82

9. The timeout for RTT3 is 676.18

QUESTION 1 OF 9

What is the estimatedRTT after the first RTT?

Answer

CHECK HINT SKIP

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